

You're in Linux... now what?

(Navigation)

Adapted from materials from Dr. Carrier

```
fergus@eos01: /  
fergus@eos01:/$ ls -la  
total 4194684  
drwxr-xr-x 27 root root 4096 Aug 15 14:21 .  
drwxr-xr-x 27 root root 4096 Aug 15 14:21 ..  
drwxr-xr-x 7 ansible ansible 4096 Aug 15 14:24 ansible  
lrwxrwxrwx 1 root root 7 Apr 22 09:08 bin -> usr/bin  
drwxr-xr-x 2 root root 4096 Feb 26 07:58 bin.usr-is-merged  
drwxr-xr-x 5 root root 4096 Aug 22 06:10 boot  
dr-xr-xr-x 2 root root 4096 Apr 24 07:28 cdrom  
drwxr-xr-x 23 root root 5300 Aug 25 16:27 dev  
drwxr-xr-x 179 root root 12288 Aug 23 06:38 etc  
drwxr-xr-x 1218 root root 126976 Aug 23 14:01 home  
drwxr-xr-x 4 root users 4096 Aug 15 14:39 lab  
lrwxrwxrwx 1 root root 7 Apr 22 09:08 lib -> usr/lib  
lrwxrwxrwx 1 root root 9 Aug 15 14:21 lib32 -> usr/lib32  
lrwxrwxrwx 1 root root 9 Apr 22 09:08 lib64 -> usr/lib64  
drwxr-xr-x 2 root root 4096 Apr 8 10:37 lib.usr-is-merged  
drwx----- 2 root root 16384 Aug 14 09:28 lost+found  
drwxr-xr-x 2 root root 4096 Apr 24 06:47 media  
drwxr-xr-x 2 root root 4096 Apr 24 06:47 mnt  
drwxr-xr-x 8 root root 4096 Aug 15 22:23 opt  
dr-xr-xr-x 388 root root 0 Aug 15 15:39 proc  
drwx----- 12 root root 4096 Aug 20 16:26 root  
drwxr-xr-x 46 root root 1460 Aug 25 16:27 run  
lrwxrwxrwx 1 root root 8 Apr 22 09:08/sbin -> usr/sbin  
drwxr-xr-x 2 root root 4096 Mar 31 05:00/sbin.usr-is-merged  
drwxr-xr-x 27 root root 4096 Aug 15 14:15 snap  
drwxr-xr-x 2 root root 4096 Apr 24 06:47 srv  
-rw----- 1 root root 4294967296 Aug 14 09:31 swap.img  
dr-xr-xr-x 13 root root 0 Aug 15 15:39 sys  
drwxrwxrwt 21 root root 4096 Aug 25 21:06 tmp  
drwxr-xr-x 13 root root 4096 Aug 15 14:21 usr  
drwxr-xr-x 15 root root 4096 Aug 15 15:39 var  
drwxrwx--x 58 root computing-faculty 4096 Jul 22 09:55 WEB_FACULTY  
drwxr-xr-x 2954 root root 73728 Feb 1 2023 WEB_STUDENT  
fergus@eos01:/$
```

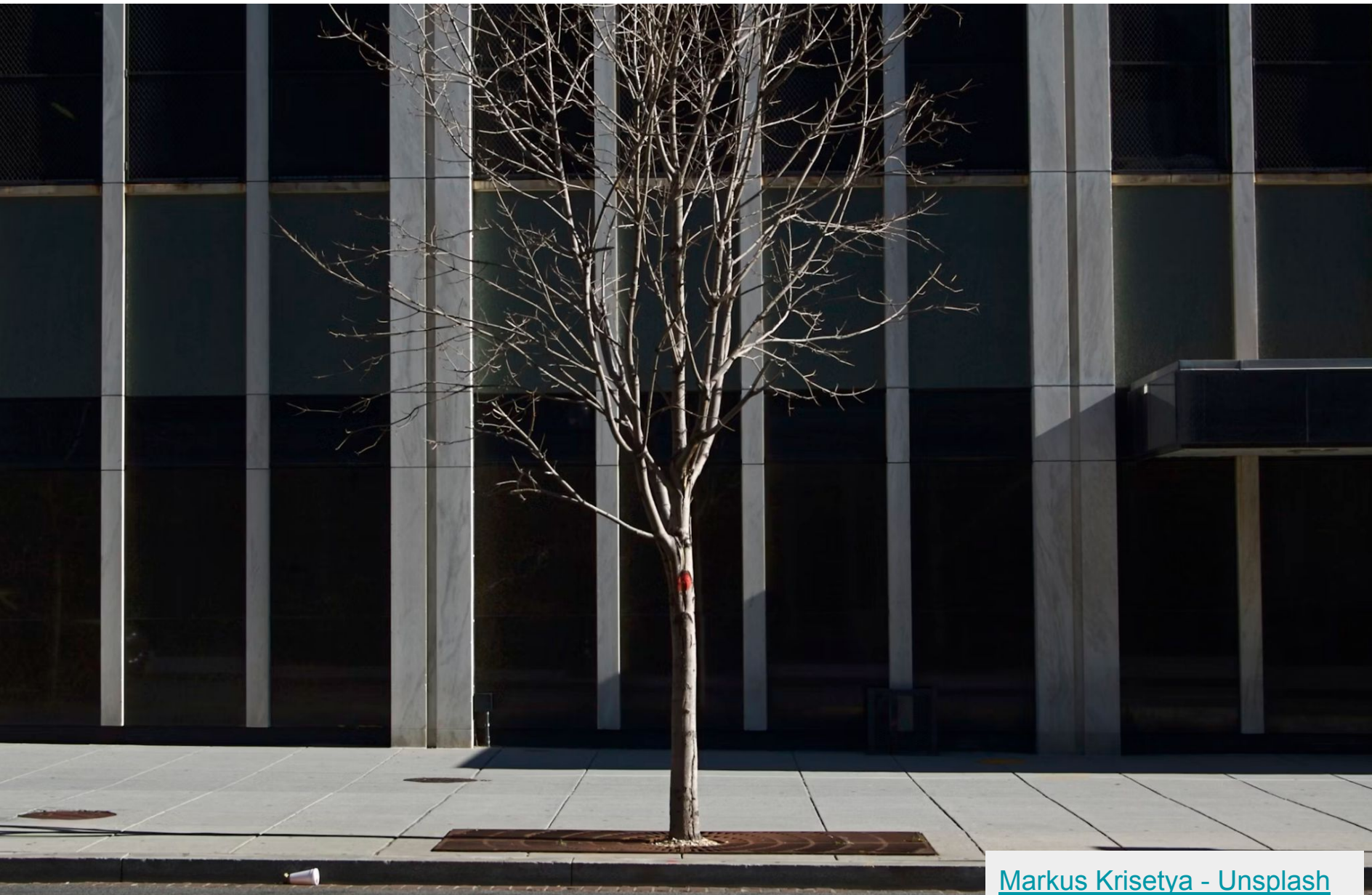
The Linux Filesystem

The Linux Filesystem

- The Linux filesystem is ***hierarchical***

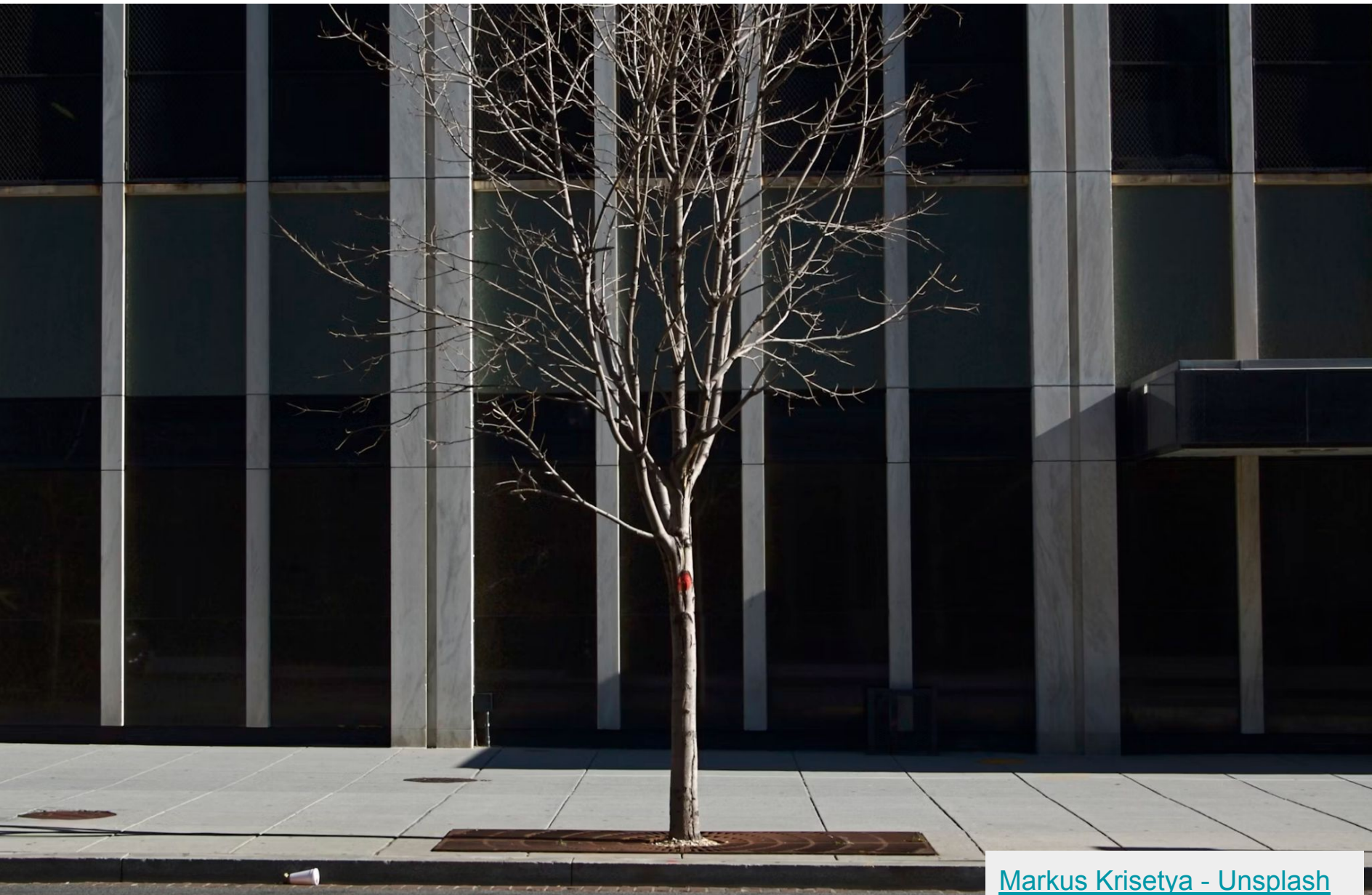
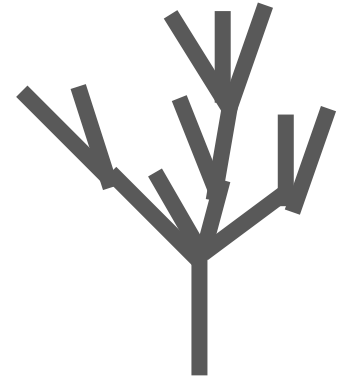
The Linux Filesystem

- The Linux filesystem is ***hierarchical***



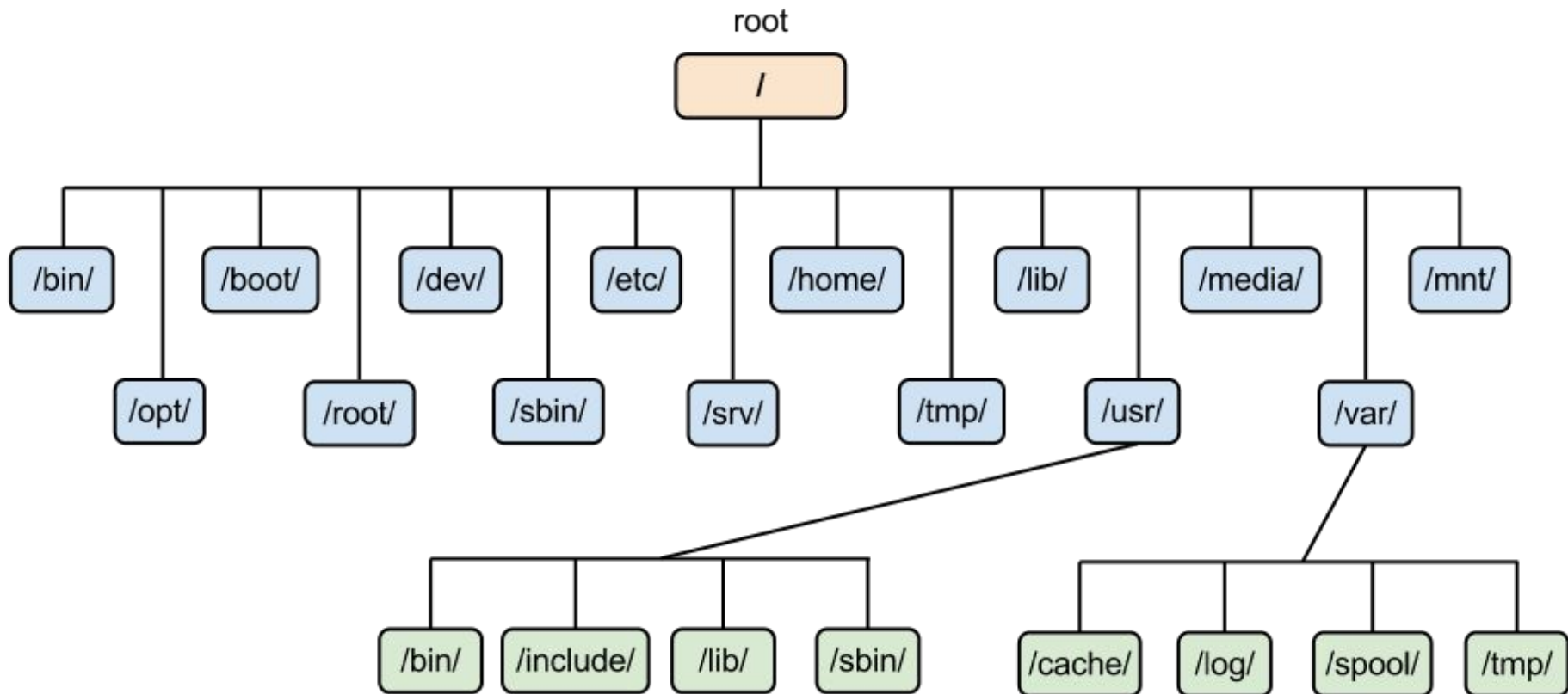
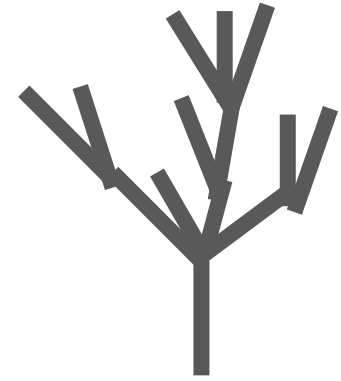
The Linux Filesystem

- The Linux filesystem is ***hierarchical***



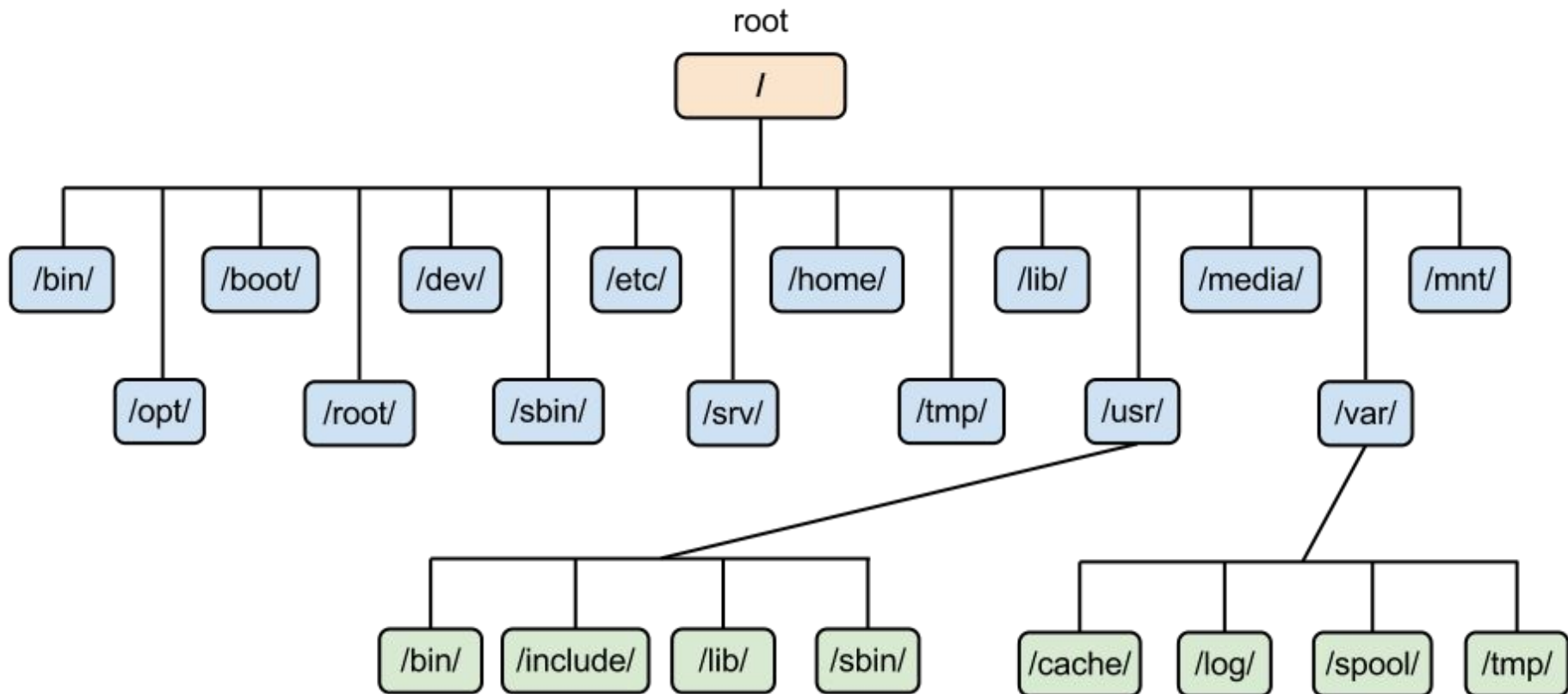
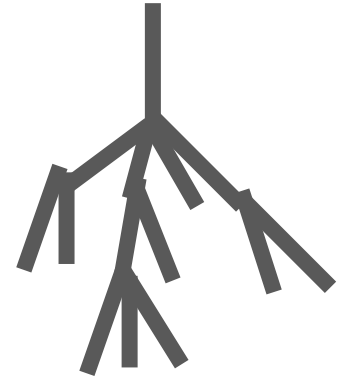
The Linux Filesystem

- The Linux filesystem is ***hierarchical***



The Linux Filesystem

- The Linux filesystem is ***hierarchical***



Important directories

Important directories

Root:

Important directories

Root: `/` (literally just a forward slash)

Important directories

Root: `/` (literally just a forward slash)

- The basis of the entire file system

Important directories

Root: `/` (literally just a forward slash)

- The basis of the entire file system

Your home directory:

`/home/username`

Important directories

Root: `/` (literally just a forward slash)

- The basis of the entire file system

Your home directory:

`/home/username`

However, we have a shortcut!

`~` (just a tilda!)

Important directories

Root: `/` (literally just a forward slash)

- The basis of the entire file system

Your home directory:

`/home/username`

However, we have a shortcut!

`~` (just a tilda!)

- The main location for your personal (user) files

Paths

Two flavors:

Paths

Two flavors:

- Absolute: Fully defined from root
 - Start with / or ~

Paths

Two flavors:

- Absolute: Fully defined from root
 - Start with / or ~
- Relative: from current directory

Basic commands

Basic commands

- `cd` - **C**hange into another **d**irectory

Basic commands

- `cd` - **C**hange into another **d**irectory
- `ls` - **L**ist files in current directory

Basic commands

- `cd` - **C**hange into another **d**irectory
- `ls` - **L**ist files in current directory
- `mkdir` - **M**ake a **d**irectory

Basic commands

- `cd` - **C**hange into another **d**irectory
- `ls` - **L**ist files in current directory
- `mkdir` - **M**ake a **d**irectory
- `rm` - **R**emove file/directory

} Be careful!

Basic commands

- `cd` - **C**hange into another **d**irectory
 - `ls` - **L**ist files in current directory
 - `mkdir` - **M**ake a **d**irectory
 - `rm` - **R**emove file/directory
 - `cp` - **C**opy
- 
- Be careful!

Basic commands

- `cd` - **C**hange into another **d**irectory
 - `ls` - **L**ist files in current directory
 - `mkdir` - **M**ake a **d**irectory
 - `rm` - **R**emove file/directory
 - `cp` - **C**opy
 - `mv` - **M**ove
- 
- Be careful!

Basic commands

- `cd` - **C**hange into another **d**irectory
 - `ls` - **L**ist files in current directory
 - `mkdir` - **M**ake a **d**irectory
 - `rm` - **R**emove file/directory
 - `cp` - **C**opy
 - `mv` - **M**ove
 - Also used to rename (move to a different name)
- 
- Be careful!

Basic commands

- `cd` - **C**hange into another **d**irectory
 - `ls` - **L**ist files in current directory
 - `mkdir` - **M**ake a **d**irectory
 - `rm` - **R**emove file/directory
 - `cp` - **C**opy
 - `mv` - **M**ove
 - Also used to rename (move to a different name)
 - `pwd` - **P**rint **w**orking **d**irectory
- 

Important directories (part 2)

`.` (just a dot) - current directory

Important directories (part 2)

 (just a dot) - current directory

 (two dots) - parent directory

Command options

- Most commands have options to modify behavior

Command options

- Most commands have options to modify behavior
 - Example:
 - `ls -la`

Command options

- Most commands have options to modify behavior
 - Example:
 - `ls -la`
 - Equivalent to `ls -l -a`

Command options

- Most commands have options to modify behavior
 - Example:
 - `ls -la`
 - Equivalent to `ls -l -a`
 - `-l` and `-a` are the options

Command options

- Most commands have options to modify behavior
 - Example:
 - `ls -la`
 - Equivalent to `ls -l -a`
 - `-l` and `-a` are the options
- How do we know how what options are available?

Command options

- Most commands have options to modify behavior
 - Example:
 - `ls -la`
 - Equivalent to `ls -l -a`
 - `-l` and `-a` are the options
- How do we know how what options are available?
 - Basic:
 - Most commands have a `--help` option (two dashes)

Command options

- Most commands have options to modify behavior
 - Example:
 - `ls -la`
 - Equivalent to `ls -l -a`
 - `-l` and `-a` are the options
- How do we know how what options are available?
 - Basic:
 - Most commands have a `--help` option (two dashes)
 - Advanced:
 - We'll talk about this soon!